Average :- Equal Distribution

like,

	A	B	C	Total
Unequal	10	5	6	$\boxed{21}$
Equal	7	7	7	$\boxed{21}$

$$\left\{ \text{Computation} \Rightarrow \frac{10+5+6}{3} = \frac{21}{3} = 7 \right\}$$

$$\boxed{\text{Average} = \frac{\text{Total Sum of Values}}{\text{Number of Values}}}$$

In short, $A = \dfrac{S}{n}$

$$\text{Sum} = \text{Average} \times \text{Number of Values}$$

$$S = nA$$

माना, सबके पास Unequal Values हैं, But अगर equal होती तो कितनी होती, ⟶ is called Average.

example,

A = 10 Chocolates
B = 20 Chocolates
C = 30 Chocolates

$$\text{Average} = \frac{\text{Total Sum of Chocolates}}{\text{Number of Persons}}$$

$$\text{Average} = \frac{10+20+30}{3} = \frac{60}{3} = 20$$

Average = 20 Chocolates.

Q. The average of 14, 22, 28, 31 and 45 is:
14, 22, 28, 31 और 45 का औसत है:
(A) 26 (B) 27 (C) 28 (D) 29 (E) None of these

$$\text{Average} = \frac{\text{Total Sum of Values}}{\text{No. of Values}}$$

$$\text{Average} = \frac{14+22+28+31+45}{5} = \frac{140}{5} = 28$$

Average = 28.

For understanding :-

$$\begin{array}{ccccc} 14 & 22 & 28 & 31 & 45 \\ \downarrow +14 & \downarrow +6 & \downarrow +0 & \downarrow -3 & \downarrow -17 \\ 28 & 28 & 28 & 28 & 28 \end{array}$$

- $14 + 6 + 0 - 3 - 17 \Rightarrow 0$ {overall change is zero}

Q. Find the average of 18, 24, 30, 12, – 6 and – 12.
18, 24, 30, 12, – 6 और – 12 का औसत ज्ञात कीजिए।
(A) 9 (B) 10 (C) 11 (D) 12 (E) None of these

$$\text{Average} = \frac{18+24+30+12-6-12}{6} = \frac{66}{6}$$

Average = 11

For understanding,

$$\begin{array}{cccccc} 18 & 24 & 30 & 12 & -6 & -12 \\ \downarrow -7 & \downarrow -13 & \downarrow -19 & \downarrow -1 & \downarrow +17 & \downarrow +23 \\ 11 & 11 & 11 & 11 & 11 & 11 \end{array}$$

- $-7 - 13 - 19 - 1 + 17 + 23 \Rightarrow 0$

Overall change from Average is always Zero.

The average of 12, 18, 24, 30, 42, 56 and x is 32. Find the value of x.
12, 18, 24, 30, 42, 56 और x का औसत 32 है। x का मान ज्ञात कीजिए।
(A) 36 (B) 40 (C) 42 (D) 44 (E) None of these

$$\frac{12+18+24+30+42+56+x}{7} = 32$$

$$182 + x = 32 \times 7$$

$$182 + x = 224$$

$$\boxed{x = 42}$$

For understanding,

12	18	24	30	42	56	x
↓	↓	↓	↓	↓	↓	↓
32	32	32	32	32	32	32
+20	+14	+8	+2	−10	−24	$\boxed{-10}$

0

$x - 10 = 32$

$\boxed{x = 42}$

The average of 8, 12, 16, 20, 24, 32, x and 2.5x is 28. Find the value of x.
8, 12, 16, 20, 24, 32, x और 2.5x का औसत 28 है। x का मान ज्ञात कीजिए।
(A) 32 (B) 26 (C) 28 (D) 30 (E) None of these

$$\frac{8+12+16+20+24+32+x+2.5x}{8} = 28$$

$$112 + 3.5x = 224$$

$$3.5x = 112$$

$$\boxed{x = 32}$$

Method 2,

8	12	16	20	24	32	x	$2.5x$
↓	↓	↓	↓	↓	↓	↓	↓
28	28	28	28	28	28	28	28
+20	+16	+12	+8	+4	−4	{−56}	

$$56 - 3.5x = -56$$

$$112 = 3.5x$$

$$\boxed{x = 32}$$

The average of (x + 6), (2x + 4), (30 – 3x), 12 and 18 is:
(x + 6), (2x + 4), (30 – 3x), 12 और 18 का औसत है:
(A) 12 (B) 14 (C) 16 (D) 18 (E) None of these

$$\text{Average} = \frac{\text{Sum of Values}}{\text{No. of Values}}$$

$$\text{Average} = \frac{(x+6)+(2x+4)+(30-3x)+12+18}{5}$$

$$\text{Average} = \frac{3x+70-3x}{5}$$

$$\text{Average} = \frac{70}{5} \Rightarrow 14$$

The average of (x + 2), (x + 6), (x + 10) and (x + 14) is:
(x + 2), (x + 6), (x + 10) और (x + 14) का औसत है:
(A) x + 6 (B) x + 8 (C) x + 10 (D) 2x + 8 (E) None of these

$$\text{Average} = \frac{(x+2)+(x+6)+(x+10)+(x+14)}{4}$$

$$\text{Average} = \frac{4x+32}{4}$$

$$\text{Average} = \frac{4(x+8)}{4}$$

$$\text{Average} = x+8$$

Understanding,

$$\begin{array}{cccc} (x+2) & (x+6) & (x+10) & (x+14) \\ \downarrow & & & \\ (x+8) & (x+8) & (x+8) & (x+8) \\ \underbrace{+6 \quad\quad\quad +2} & & & \end{array}$$

$$\underbrace{+6 \qquad +2 \qquad -2 \qquad -6}_{0}$$

A total amount of Rs. 648 was distributed among a certain number of children. If the average amount received by each child was Rs. 18, find the number of children.
कुल Rs. 648 की राशि कुछ बच्चों में बाँटी गई। यदि प्रत्येक बच्चे को प्राप्त औसत राशि Rs. 18 थी, तो बच्चों की संख्या ज्ञात कीजिए।
(A) 32 (B) 36 (C) 40 (D) 28 (E) None of these

$$\text{Average} = \frac{\text{Total Amount}}{\text{No. of Children}}$$

$$18 = \frac{648}{n}$$

$$n = \frac{648}{18}$$

$n = 36$ children

A total of 1080 chocolates are distributed equally among x students, so that each student gets 36 chocolates. If the same chocolates are distributed equally among (x + 10) students, how many chocolates will each student get?
कुल 1080 चॉकलेट x छात्रों में समान रूप से बाँटी जाती हैं, जिससे प्रत्येक छात्र को 36 चॉकलेट मिलती हैं। यदि वही चॉकलेट (x + 10) छात्रों में समान रूप से बाँटी जाएँ, तो प्रत्येक छात्र को कितनी चॉकलेट मिलेंगी?
(A) 24 (B) 27 (C) 30 (D) 32 (E) None of these

$$\text{Average} = \frac{\text{Total Chocolates}}{\text{No. of students } (x)}$$

$$36 = \frac{1080}{x}$$

$$x = \frac{1080}{36} \Rightarrow \boxed{x = 30}$$

$$\boxed{x + 10 = 40}$$

$\Rightarrow$ In 40 students,

$$\text{Average} = \frac{1080 \text{ chocolates}}{40 \text{ students}}$$

Average = 27 chocolates.

The same amount of money is distributed among the members of Group A and Group B separately. Each member of Group A gets an average of Rs. 50, while each member of Group B gets an average of Rs. 45. Find the ratio of the number of members in Group A to Group B.
समान राशि Group A और Group B के सदस्यों में अलग-अलग बाँटी जाती है। Group A के प्रत्येक सदस्य को औसतन Rs. 50 मिलते हैं, जबकि Group B के प्रत्येक सदस्य को औसतन Rs. 45 मिलते हैं। Group A और Group B के सदस्यों की संख्या का अनुपात ज्ञात कीजिए।

(A) 3 : 7 (B) 4 : 10 (C) 5 : 4 (D) 9 : 10 (E) None of these

Let Total Amount to be distributed = S

$\text{Average} = \frac{\text{Sum}}{\text{No.}}$, So, $\boxed{\text{No. of Members} = \frac{\text{Sum}}{\text{Average}}}$

Group A Group B

$n \Rightarrow \frac{S}{50} \qquad \frac{S}{45}$

$\cancel{S} \times 45 : \cancel{S} \times 50$

$45 : 50$

$9 : 10$

if a sum is distributed among '2x' students then each one will get an average amount of Rs. 72. What will be the amount with each student, if the same sum is distributed among '3x' students?
यदि कोई राशि '2x' छात्रों में बाँटी जाए, तो प्रत्येक को औसतन Rs. 72 मिलेंगे। यदि वही राशि '3x' छात्रों में बाँटी जाए, तो प्रत्येक छात्र को कितनी राशि मिलेगी?

(A) 48 (B) 36 (C) 42 (D) 54 (E) None of these

Average = Rs. 72

No. of Students = 2x

Sum = Rs. 72 × 2x

= 144x

\# Distribution to (3x) students :-

$\text{Average} = \frac{\text{Sum}}{\text{Students}}$

$\text{Average} = \frac{144\cancel{x}}{3\cancel{x}} \Rightarrow \boxed{\text{Average} = 48}$

Properties of Average :-

1) If each observation is increased or decreased by n, then Average will also increase or Decrease by n.

$$\boxed{12} \xrightarrow{+5} \boxed{17} \quad \boxed{15} \xrightarrow{+5} \boxed{20} \quad \boxed{17} \xrightarrow{+5} \boxed{22} \quad \boxed{24} \xrightarrow{+5} \boxed{29}$$

$$\text{Avg.} = \boxed{17} \xrightarrow{+5} \boxed{22}$$

2) If each observation is multiplied or divided by n, then Average will also be multiplied or divided by n.

$$\boxed{2} \xrightarrow{\times 3} \boxed{6} \quad \boxed{7} \xrightarrow{\times 3} \boxed{21} \quad \boxed{11} \xrightarrow{\times 3} \boxed{33} \quad \boxed{20} \xrightarrow{\times 3} \boxed{60} \quad \boxed{25} \xrightarrow{\times 3} \boxed{75}$$

$$\text{Avg.} = \boxed{13} \xrightarrow{\times 3} \boxed{39}$$

Result :- Any kind of same operation that involves $+, -, \times, \div$, if applied on each observation, it results in the same change with Average.

3) If Sum increases by x, then Average increases by $\frac{x}{n}$. (n is same)

$$A = \frac{S}{n}$$

$$\frac{S+x}{n} \longrightarrow \frac{S}{n} + \frac{x}{n} \longrightarrow A + \frac{x}{n}$$

example,

Sum = 450

n = 5 {Sum increases by 15}

Average = $\frac{\text{Sum}}{n}$

Average = $\frac{450}{5} \Rightarrow 90$

* Sum increases by 15 {New sum = 465}

Average = $\frac{465}{5} \Rightarrow 93$

In short,

it is increased by 3 {i.e., $\frac{\text{Change}}{n} = \frac{15}{5} = 3$}

4) If sum decreases by x, then Average also decreases by $\frac{x}{n}$ (n is same)

$$A = \frac{S}{n}$$

$$\frac{S-x}{n} \longrightarrow \frac{S}{n} - \frac{x}{n} \longrightarrow A - \frac{x}{n}$$

example,

Sum = 343 {Sum decreases by 35} New Sum = 308

n = 7 n = 7

Average = $\frac{343}{7} \Rightarrow 49$

$\Downarrow$

Avg. decreased by $\frac{35}{7} \Rightarrow 5$

Average = $\frac{308}{7} \Rightarrow 44$

-5

The present ages of six persons are 12 years, 14 years, 18 years, 22 years, 31 years and 35 years. What will be their average age after 5 years?
छह व्यक्तियों की वर्तमान आयु 12 वर्ष, 14 वर्ष, 18 वर्ष, 22 वर्ष, 31 वर्ष और 35 वर्ष है। 5 वर्ष बाद उनकी औसत आयु क्या होगी?

(A) 28 years (B) 29 years (C) 27 years (D) 25 years (E) None of these

$\boxed{12} \xrightarrow{+5}$ $\boxed{14} \xrightarrow{+5}$ $\boxed{18} \xrightarrow{+5}$ $\boxed{22} \xrightarrow{+5}$ $\boxed{31} \xrightarrow{+5}$ $\boxed{35} \xrightarrow{+5}$

$$\text{Average} = \frac{12+14+18+22+31+35}{6}$$

$$= \frac{132}{6} \Rightarrow \boxed{22} \xrightarrow{+5} \boxed{27}$$

* If all observations increase by 5, then Average will also increase by 5.

The average weight of 54 persons is 47 kg. All of them gained 2.5 kg and then each one of them lost 8 kg. What was their final average weight?
54 व्यक्तियों का औसत वजन 47 kg है। सभी का वजन 2.5 kg बढ़ गया और फिर प्रत्येक का 8 kg कम हो गया। उनका अंतिम औसत वजन क्या था?

(A) 42 kg (B) 42.5 kg (C) 40.5 kg (D) 41.5 kg (E) None of these

All persons gained 2.5 kg and all persons lost 8 kg.

Average ⇒ 47 kg.
\+ Gained +2.5 kg
− Lost − 8 kg

New Average ⇒ 41.5 kg

The average quantity of liquid in 175 packages is 24 litres. If the quantity of liquid in each package becomes 1.5 times, what will be the new average quantity per package?
175 पैकेटों में द्रव की औसत मात्रा 24 litres है। यदि प्रत्येक पैकेट में द्रव की मात्रा 1.5 गुना हो जाए, तो प्रति पैकेट नई औसत मात्रा क्या होगी?

(A) 30 litres (B) 34 litres (C) 36 litres (D) 38 litres (E) None of these

1 2 3 4 5 - - - - - - - - 174 175

$\downarrow \times 1.5$ $\downarrow \times 1.5$ $\downarrow \times 1.5$ $\downarrow \times 1.5$ $\downarrow \times 1.5$ $\downarrow \times 1.5$ $\downarrow \times 1.5$

Quantity in each package is 1.5 times, so the New Average will also be 1.5 times of the initial Average.

Average $\Rightarrow$ 24 litres

$\downarrow \times 1.5$

New Average $\Rightarrow$ 36 litres

The average length of 120 wooden planks is 72 metres. If each plank is cut into 3 equal parts, what will be the average length of each new part?
120 लकड़ी के तख्तों की औसत लंबाई 72 metres है। यदि प्रत्येक तख्ते को 3 समान भागों में काट दिया जाए, तो प्रत्येक नए भाग की औसत लंबाई क्या होगी?

(A) 18 metres (B) 20 metres (C) 22 metres (D) 24 metres (E) None of these

Total Wooden Planks = 120

* If each wooden plant is cut into 3 equal parts, (means divided by 3), then Average length will also be divided by 3.

Average $\Rightarrow$ 72 metres

$\downarrow \div 3$

New Average $\Rightarrow$ 24 metres

The average price of 160 items was Rs. 450. The price of each article was first increased by 20% and then each article was sold at 1/5th of its increased price. What was the average selling price of each item?
160 वस्तुओं का औसत मूल्य Rs. 450 था। प्रत्येक वस्तु का मूल्य पहले 20% बढ़ाया गया और फिर प्रत्येक वस्तु को उसके बढ़े हुए मूल्य के 1/5 भाग पर बेचा गया। प्रत्येक वस्तु का औसत विक्रय मूल्य क्या था?

(A) Rs. 98 (B) Rs. 108 (C) Rs. 112 (D) Rs. 118 (E) None of these

increased by 20% $\Rightarrow 1+\frac{1}{5} \Rightarrow \frac{6}{5}$

* New Average $\Rightarrow \left\{ \overset{18}{\cancel{450}} \times \frac{6}{\cancel{5}} \right\} \times \frac{1}{\cancel{5}} \Rightarrow$ Rs. 108

The average salary of 154 employees is Rs. 45,000. Each employee gets an increment of 10% and then a deduction of Rs. 2,500 is made from the salary. What will be the new average in-hand salary per employee?
154 कर्मचारियों का औसत वेतन Rs. 45,000 है। प्रत्येक कर्मचारी को 10% की वृद्धि मिलती है और फिर वेतन में से Rs. 2,500 की कटौती की जाती है। प्रति कर्मचारी नया औसत हाथ में मिलने वाला वेतन क्या होगा?

(A) Rs. 46,000 (B) Rs. 47,000 (C) Rs. 47,500 (D) Rs. 48,000 (E) None of these

increment of 10% $\Rightarrow 1+\frac{1}{10} \Rightarrow \frac{11}{10}$

New Average Salary $\Rightarrow \left\{ 4500\cancel{0} \times \frac{11}{1\cancel{0}} \right\} - 2500$

$\Rightarrow$ Rs. 47000

A group has 20 members with an average weight of 55 kg. Out of them, 12 members lose 2 kg each and the remaining members gain 5 kg each. Find the final average weight of the group.
एक समूह में 20 सदस्य हैं जिनका औसत वजन 55 kg है। उनमें से 12 सदस्यों का वजन 2 kg कम हो जाता है और शेष सदस्यों का वजन 5 kg बढ़ जाता है। समूह का अंतिम औसत वजन ज्ञात कीजिए।

(A) 55.8 kg (B) 56.2 kg (C) 56.6 kg (D) 57 kg (E) None of these

Total members $\Rightarrow$ 20

12 members $\times (-2)$ 8 members $\times (+5)$

$\underbrace{-24 \qquad\qquad +40}_{+16}$

Change in Average $\Rightarrow \frac{16}{20} \Rightarrow + 0.8$ kg

New Average $\Rightarrow$ 55 kg + 0.8 kg $\Rightarrow$ 55.8 kg

The average weight of 20 boys in a class is 58 kg and the average weight of 30 girls is 52 kg. Find the average weight of all the students together.

एक कक्षा में 20 लड़कों का औसत वजन 58 kg है और 30 लड़कियों का औसत वजन 52 kg है। सभी विद्यार्थियों का संयुक्त औसत वजन ज्ञात कीजिए।

(A) 54.4 kg (B) 54.2 kg (C) 53.8 kg (D) 54.8 kg (E) None of these

	Boys	Girls	
Average	58 kg	52 kg	
No. of –	20	30	(2 : 3)
Sum ⇒	(58 x 20) = 1160	(52 x 30) = 1560	

$$\text{Average} = \frac{\text{Sum}}{n} \Rightarrow \frac{1160 + 1560}{50} \Rightarrow 54.4 \text{ kg.}$$

Through Weighted Average,

Average	58	52
No. (Ratio)	2 :	3

$$\text{Average}\ \{\text{overall}\} \Rightarrow \frac{(58 \times 2) + (52 \times 3)}{5} \Rightarrow \frac{272}{5} \Rightarrow 54.4 \text{ kg.}$$

\# Weighted Average ⇒

While calculating Average, Do not take exact values of n, take their Ratios instead. (for easy calculation)

* $\boxed{\text{Weighted} \equiv \text{Ratio}}$